What is Batch Norm.

Wednesday, March 25, 2026 7:43 PM

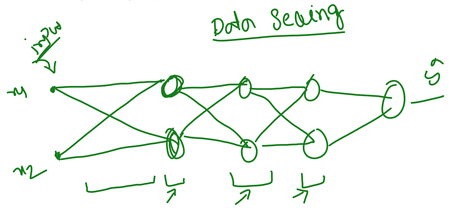
What is Batch Normalization?

Batch Normalization (BN) is a technique designed to make the training of Deep Neural Networks (DNNs) faster and more stable.

It works by normalizing the activation values of hidden layers using the mean and variance computed from the current mini-batch. This normalization is typically applied just before or after the activation (nonlinear) function.

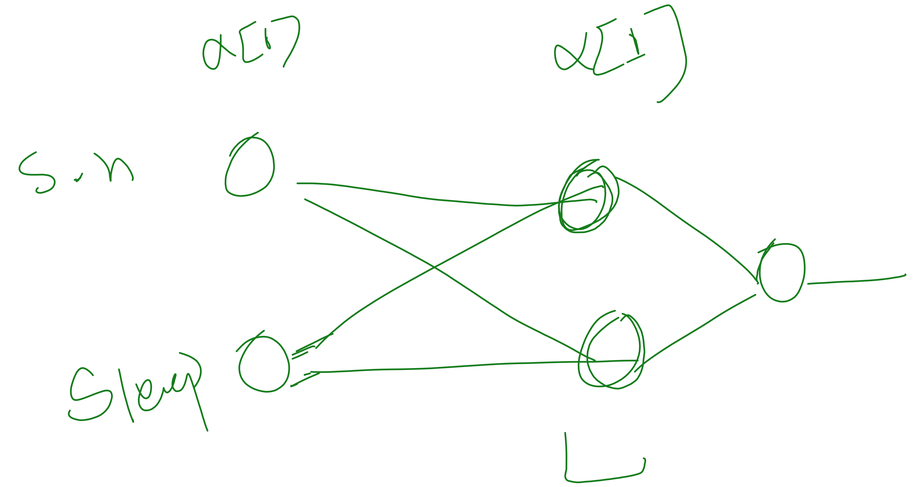
Normalization

Data Scaling



why Batch Normalization

Study hours	Sleep	CGPA
2	3	3.2
5	7	3.1
8	9	3.5
10	11	3.6
13	14	3.7



input normalized

$\mu = 0$ $\sigma = 1$

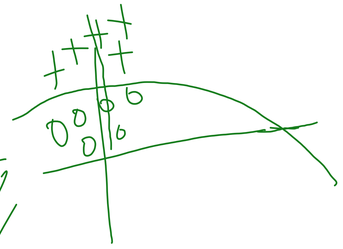
mean

std

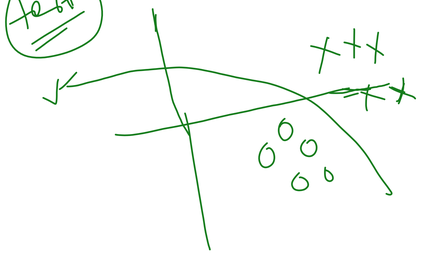
Covariate Shift

Rose = y=1
y=0

training sample



test



- What is **Data Distribution**?

Data distribution describes **how values in a dataset are spread or arranged**.

It answers:

- What values appear?
- How often do they appear?
-

What happens if input distribution changes after training?

When your model is trained, it **learns patterns based on the training data distribution**.

If the input distribution changes during testing or real-world use:

1. Performance drops

- The model sees **unfamiliar patterns**
- Predictions become **less accurate**
- Test accuracy can fall significantly

2. Features no longer align

Red Rose

Rose

Example:

- During training:
  - Age range: 20–40
- During testing:
  - Age range: 60–80

The model hasn't learned these patterns → poor predictions

3. Decision boundary becomes wrong
- The model's learned boundaries are based on old data

Train:
. Bright, high-quality images
Test:
. Dark, blurry images
Same object, but different distribution → model struggles

Definition of Internal Covariate Shift

Internal covariate shift refers to the phenomenon where the **distribution of inputs to a layer in a neural network changes during training**, as the parameters of the previous layers are updated.

In Simple Words

- Each layer depends on the output of the previous layer
- But as training goes on, those outputs **keep changing**

  👉 So every layer has to **constantly adapt to new input distributions**

📌 Why it happens

During training:
- We update weights using gradient descent
- This changes the output (activations) of each layer
- So the **input distribution of the next layer keeps shifting**

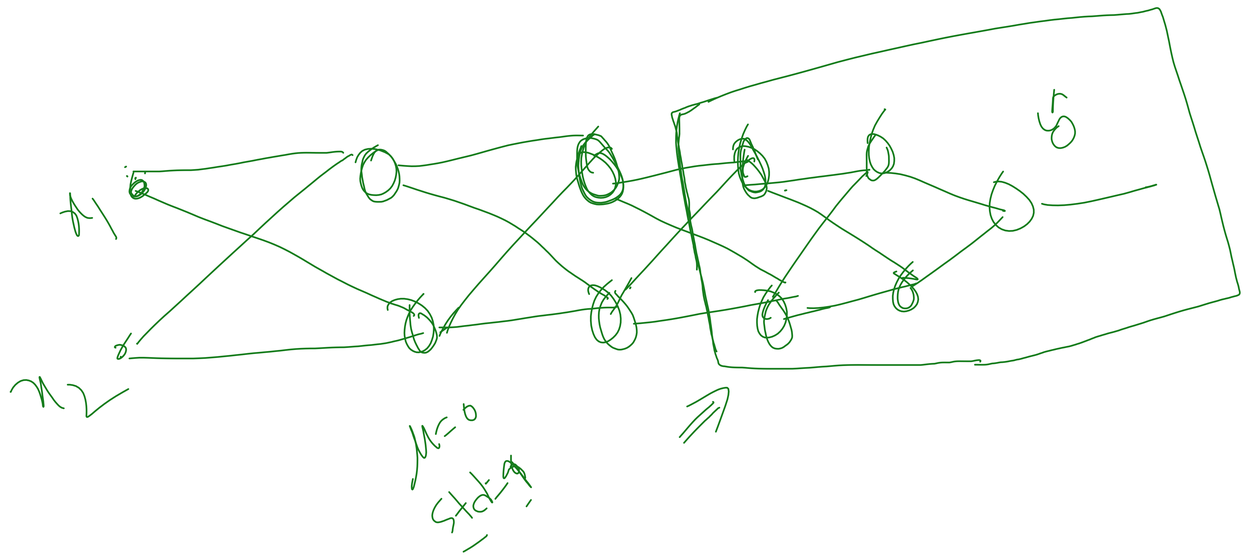
Why it's a problem

- Slows down training
- Makes learning unstable
- Requires smaller learning rates
- Harder for deeper networks to converge

One-Line Summary

Internal covariate shift =

“The continuous change in the distribution of layer inputs during training.”



Batch normalization

(4,2)

4

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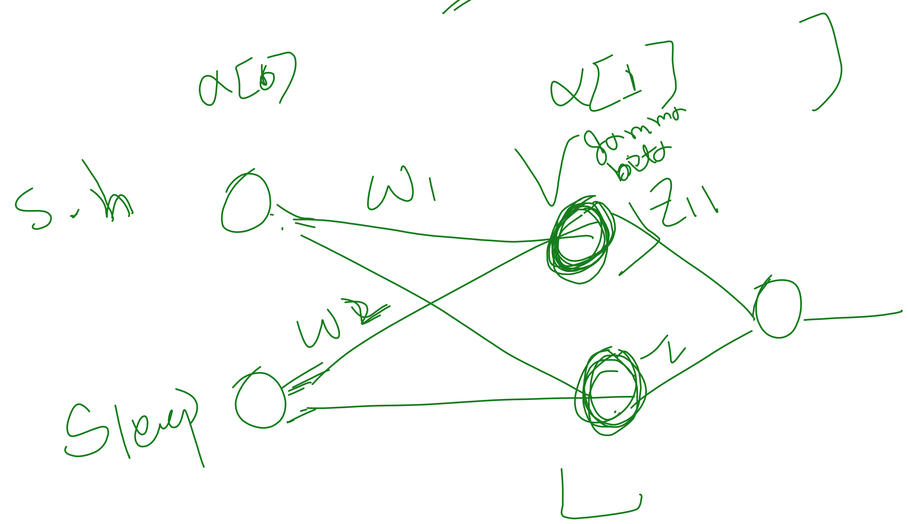

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(0) mini batch G.D

(2) layer by layer

$$Z_{11} = w_1 \times S.n + w_2 \times Sleep + b$$

normal flow :

$$g(Z_{11}) \rightarrow a_{11}$$

mean = 0

std = 1

B.N :

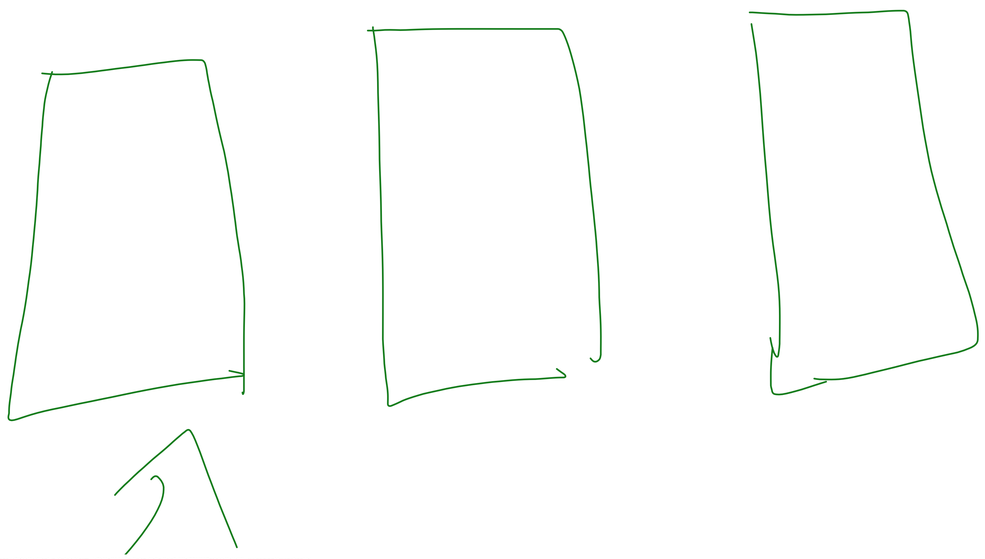
with flow of

$$\boxed{z_{11} \rightarrow z_{11}^{N} \rightarrow g(z_{11}^{N}) \rightarrow \alpha_{11}}$$

(1)

$$z_{11} \rightarrow g(z_{11}) \rightarrow \alpha_{11} \rightarrow \alpha_{11}^{N}$$

(2)



linear

$\boxed{\text{BatchNorm1d}}$

$$z_{11}^{N} \rightarrow \frac{z_{11} - \mu}{\sigma} \rightarrow \text{std}$$

mean $\mu \rightarrow ?$

std $\rightarrow ?$

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Shape:

$$(4,2) \times (2,2) + (1,2)$$

$$= \underline{(4,2)}$$

equation:

$$\mu_B = \frac{1}{m} \sum_{i=1}^{m} z_{ij}^{i} \qquad m=4$$

std

$$\sigma_B = \sqrt{\frac{1}{m} \sum_{i=1}^{m} (z_{ij}^{i} - \mu_B)}$$

$$\boxed{z_{11}' = \frac{z_{11}' - \mu_{MB}}{\sigma_B + \epsilon \rightarrow \text{error}}}$$

Step 01 $\rightarrow$

Step 02:

Last step:

$$\boxed{\text{Scaling \& shifting}}$$

$$z_{11}^{N} = \gamma z_{11}^{N} + \beta$$

$\gamma \rightarrow$ gamma

$\beta \rightarrow$ beta

$\gamma = 1$

$\beta = 0$

trainable parameter:

Lecture

$$z_{11} \rightarrow \overset{N}{z}_{11} \rightarrow \overset{BN}{z}_{11} \rightarrow g(\overset{BN}{z}_{11}) \rightarrow o_{11}$$